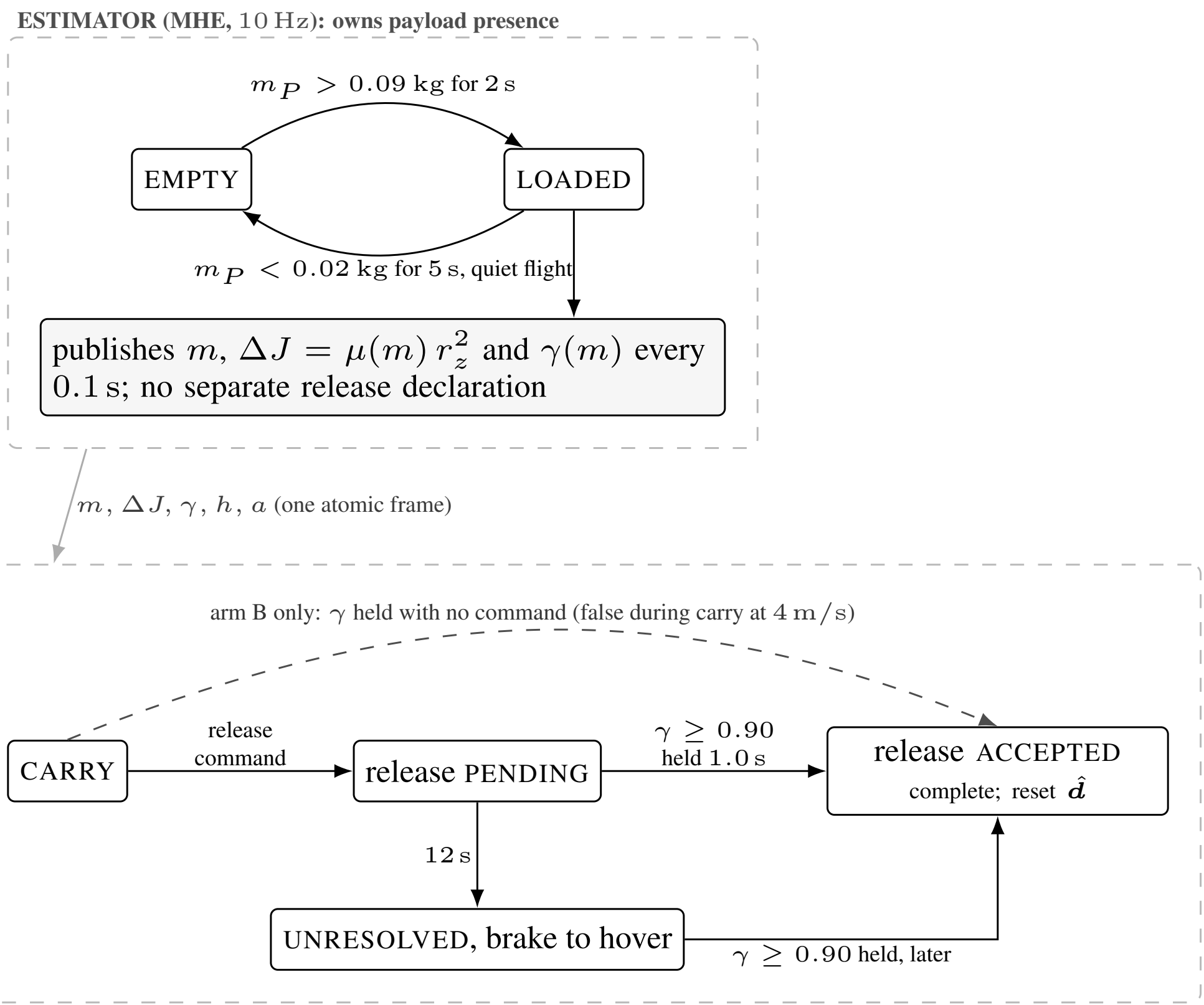

ESTIMATOR (MHE, 10 Hz): owns payload presence
$m_P > 0.09$ kg for 2 s
EMPTY
LOADED
$m_P < 0.02$ kg for 5 s, quiet flight
publishes m, $\Delta J = \mu(m)\, r_z^2$ and $\gamma(m)$ every 0.1 s; no separate release declaration
m, ΔJ, γ, h, a (one atomic frame)
arm B only: γ held with no command (false during carry at 4 m/s)
CARRY
release command
release PENDING
$\gamma \geq 0.90$ held 1.0 s
release ACCEPTED
complete; reset $\hat{\boldsymbol{d}}$
12 s
UNRESOLVED, brake to hover
$\gamma \geq 0.90$ held, later
CONTROLLER (NMPC, 20 Hz): owns the release lifecycle; m, ΔJ follow every frame regardless of state